Ishtiyak Ahmed

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Education

University of Toronto Expected May 2029
HBSc. Major in Applied Statistics; Minors in Computer Science & Geospatial Data Science

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript
Data & Analytics: SQLite, Power BI, DAX, Power Query, Tableau
Frameworks & Tools: React, Git, GitHub, Vercel
Engineering: ETL Pipelines, Data Modeling, Data Validation, Automated Testing

Experience

Research and Project Support Intern Jan. 2026 – Apr. 2026
E. Aferi *Toronto, ON*

- Conducted **one-on-one client discovery** and reviewed existing documentation to translate website requirements into a structured landing-page plan, including content organization and layout recommendations.
- Researched and contacted prospective manufacturers, collecting quotations and **comparing pricing, minimum order quantities, materials, and production timelines** against project requirements to identify sourcing constraints.
- Collected and categorized website and social-media reference materials for **approximately 38 competitors**, organizing visual research into structured folders for project review.

Projects

TTC Delay Patterns — Data Engineering & Analytics | Live Project | GitHub
Python, SQL, SQLite, Power BI, DAX, React, TypeScript

- Architected a **reproducible ETL pipeline** processing **176,004 TTC incident records** across four transit modes, integrating Python ingestion, lossless SQLite staging, validated transformations, and SQL analytical datasets.
- Engineered **idempotent ingestion** with SHA-256-verified snapshots, version-aware storage, and **transactional recovery**; tested safe reruns and interrupted loads to verify historical data preservation.
- Increased accepted incident-code description coverage from **75.97% to 93.82%**, enriching **31,411 additional records** through mode-specific dictionary matching, controlled encoding repairs, and explicit validation rules.
- Developed reconciled SQL analytical fact tables with defined data grains and safeguards against double-counting, revealing that **4.12% of Bus incidents contributed 39.46%** of recorded Bus delay minutes.
- Designed a **Power BI semantic model** with 10 tables, 13 relationships, and 54 DAX measures; published the methodology, verified findings, and limitations through a **deployed React/TypeScript case-study website**.

JavaFX Paint Editor | *Java, JavaFX, MVC, Git*

- Developed a JavaFX drawing application with **interactive shape creation**, mouse-driven editing, and selectable drawing modes using object-oriented design and MVC architecture.
- Refactored JavaFX mouse-event handling using the **Strategy design pattern**, isolating press, drag, and release behavior into **interchangeable shape manipulators** to simplify adding new drawing tools.

Tree-Based Data Visualizer | *Python, Pygame*

- Implemented a **recursive treemap algorithm** that partitions 2D display space into proportional rectangles based on hierarchical data sizes, supporting dynamic visualization of nested tree structures.
- Developed **tree mutation operations** for moving, deleting, and resizing nodes, maintaining parent-child relationships and recalculating aggregate sizes to keep visual layouts consistent after changes.
- Applied **inheritance and recursive processing** to build reusable tree models for filesystem directories and CSV datasets, supporting hierarchical visualization of file sizes and citation counts.